Adrian Cuyugan 2014-91450 Stat 266 One-Sample Wilcoxon Signed Rank Test

5.11 - The conclusion in Problem 5.10 was that the median difference (Before minus After) was positive for the affected age group, but this does imply that the reduction was the result of raising the minimum legal drinking age. Other factors, countermeasures, or advertising campaigns [like MADD (Mothers Against Drunk Drivers)] may have affected the fatality ratios. In order to investigate further, these researchers corresponding difference ratios for the 25-29 age group, who were not affected by the law change. Carry out an appropriate test and write a report of your conclusions.

State n = 10	Affected Age Group	25-29 Age Group	$D_i = X_i - Y_i$	$\lvert D_i \rvert$	r_i	Z_i
Florida	0.600	-0.025	0.625	0.625	9	1
Georgia	0.068	-0.230	0.298	0.298	8	1
Illinois	0.025	0.004	0.021	0.021	1	1
Iowa	0.078	-0.008	0.086	0.086	6.5	1
Maine	-0.022	0.061	-0.083	0.083	5	0
Michigan	0.072	0.015	0.057	0.057	2	1
Montana	0.041	-0.035	0.076	0.076	4	1
Nebraska	0.860	-0.016	0.876	0.876	10	1
New Hampshire	0.012	-0.061	0.073	0.073	3	1
Tennessee	0.035	-0.051	0.086	0.086	6.5	1

This tests for 0 difference between Affected Age Group with the 25-29 Age Group. So, the hypothesis is set as follows:

$$H_0: M = 0$$
$$H_A: M \neq 0$$

The midrank method is used to assign for tied ranks. The test statistic, T^-, is computed as follows with its relationship with $T^+ - T^-$:

$$Z_i = \begin{cases} 1 \ \ if \ \ D_i > 0 \\ 0 \ \ if \ \ D_i \leq 0 \end{cases} \qquad T^+ = \sum_{i=1}^{10} Z_i r(|D_i|) = 50 \qquad T^- = \sum_{i=1}^{10} 1 - Z_i r(|D_i|) = 5 \qquad T^+ - T^- = 2\sum_{i=1}^{10} Z_i r(|D_i|) - \frac{10(10+1)}{2} = 55$$

From Table H, the rejection region consists of the rejection for $T^-_{0.025} \leq 8$ or $T^-_{0.975} \geq 47$ at 0.05 level of significance, with exact tail probability of 0.048. Since $T^-_{computed} = 5$, the null hypothesis is rejected. Using p-value method:

$$P[T^- \leq 5 | H_0 \ is \ true]; \ P[T^- \geq 5 | H_0 \ is \ true] \qquad 2 * \text{tailed p-value} = 0.020$$

As a take away, there is enough statistical evidence that the ratio of the number of single-vehicle nighttime fatalities to the number of licensed drivers of the affected age group, even after raising the minimum drinking age, that it is not equal to the 25-29 age group.

```r
# One-Sample Wilcox Signed Rank Test
# Exact p-value may be different because this does not consider midranks for ties
affected.agegroup <- c(0.600, 0.068, 0.025, 0.078, -0.022, 0.072, 0.041, 0.860, 0.012, 0.035)
agegroup.25to29 <- c(-0.025, -0.023, 0.004, -0.008, 0.061, 0.015, -0.035, -0.016, -0.061, 0.051)

wilcox.test(x = affected.agegroup, y = agegroup.25to29,
            mu = 0, # hypothesis is that the median is equals 0
            alternative = 'two.sided', # two-tailed test
            paired = T, # one-sample paired test
            exact = T, # compute for exact probability
            correct = F, # not using continuity correction and normal approximation
            conf.int = T, conf.level = 0.95) # computes for confidence interval at 95%

##
##  Wilcoxon signed rank test
##
## data:  affected.agegroup and agegroup.25to29
## V = 48, p-value = 0.03711
## alternative hypothesis: true location shift is not equal to 0
## 95 percent confidence interval:
##  0.0025 0.4485
## sample estimates:
## (pseudo)median
##         0.0745
```